Voice Controlled Bot

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My Story

The way I came to encounter the UTSA Open Cloud Institute for research was through my mom who works at UTSA and helped me get into a program for computer science and cybersecurity under Dr. Paul Rad. Thus, I got the opportunity to work with autonomous vehicles and voice control systems.

The Project

Can an autonomous vehicle and other vehicles connect with a voice controlled system?

The Goal

Our goal was to set up a Alexa voice system to control a car.

Materials

- Donkey car website
- Alexa
- Siri
- Voice Control and Motor Control Software

Procedure

1. First we got the car to respond to computer code by finding the specified data needed.

2. Then we programmed the car to follow basic commands such as left, right, forward, and backward.

3. Then we connected the car to the voice system

What happened vs. what was planned

We thought we could use Alexa voice system functions in Amazon Web Services (AWS) to control the car.

Instead, we programmed the cars steering and throttle to work along with Siri in order to execute certain commands.

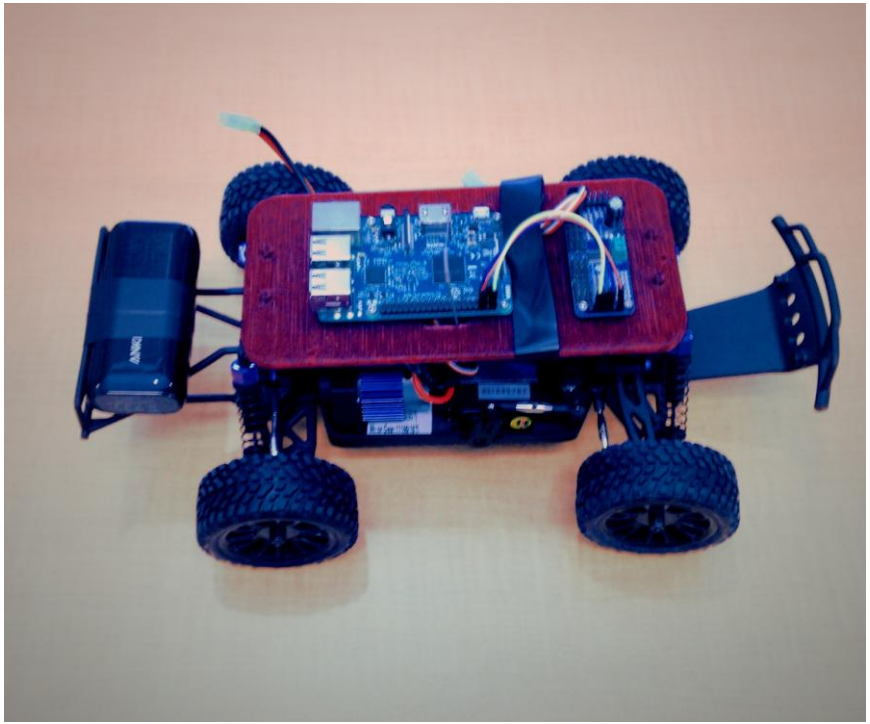
Problems we encountered

1. The throttle would not respond to PCA9685 control board commands.
   - We tested this by checking if the throttle commands would work with the remote the car came with.
   - Then we found that the command PCA9685(Channel) was used to switch between steering and the throttle and fixed the problem

2. Alexa voice system required an account and other unnecessary things.
   - We found a way to customize commands in Siri and decided that we could use this to our advantage when working with the car and changed to a Siri controlled system.

Problems we encountered(cont.)

3. Some of the code with the modules for the Siri control system didn’t connect with the throttle and steering circuit board.
    - We fixed this issue by finding a method to bring the code of the steering into the module system.



- Sign up for Camps
- Network with teachers to get connections
- Choose something you like to do
- Talk to teachers and parents
- Find people who are experienced and can teach you

Advice

CONCLUSION

An Alexa voice system can be used to control a car, but using a Siri voice system was more efficient.

Special Thanks

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